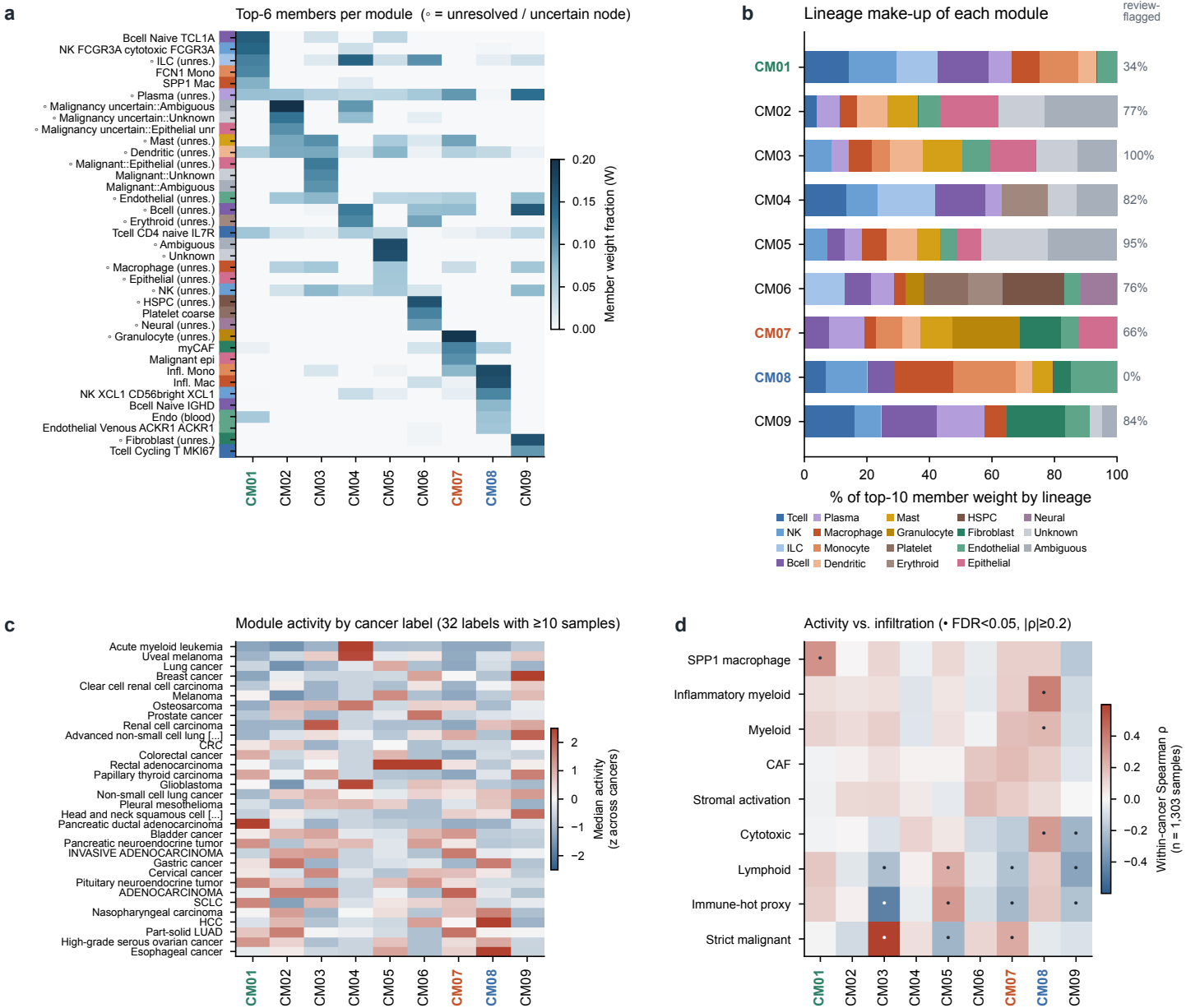

a
Top-6 members per module (◦ = unresolved / uncertain node)
Bcell Naive TCL1A
NK FCGR3A cytotoxic FCGR3A
◦ ILC (unres.)
FCN1 Mono
SPP1 Mac
◦ Plasma (unres.)
◦ Malignancy uncertain::Ambiguous
◦ Malignancy uncertain::Unknown
◦ Malignancy uncertain::Epithelial unr
◦ Mast (unres.)
◦ Dendritic (unres.)
◦ Malignant::Epithelial (unres.)
Malignant::Unknown
Malignant::Ambiguous
◦ Endothelial (unres.)
◦ Bcell (unres.)
◦ Erythroid (unres.)
Tcell CD4 naive IL7R
◦ Ambiguous
◦ Unknown
◦ Macrophage (unres.)
◦ Epithelial (unres.)
◦ NK (unres.)
◦ HSPC (unres.)
Platelet coarse
◦ Neural (unres.)
◦ Granulocyte (unres.)
myCAF
Malignant epi
Infl. Mono
Infl. Mac
NK XCL1 CD56bright XCL1
Bcell Naive IGHD
Endo (blood)
Endothelial Venous ACKR1 ACKR1
◦ Fibroblast (unres.)
Tcell Cycling T MKI67
0.20
0.15
0.10
0.05
0.00
Member weight fraction (W)
CM01 CM02 CM03 CM04 CM05 CM06 CM07 CM08 CM09
b
Lineage make-up of each module
review-flagged
CM01 34%
CM02 77%
CM03 100%
CM04 82%
CM05 95%
CM06 76%
CM07 66%
CM08 0%
CM09 84%
0 20 40 60 80 100
% of top-10 member weight by lineage
Tcell Plasma Mast HSPC Neural
NK Macrophage Granulocyte Fibroblast Unknown
ILC Monocyte Platelet Endothelial Ambiguous
Bcell Dendritic Erythroid Epithelial
c
Module activity by cancer label (32 labels with ≥10 samples)
Acute myeloid leukemia
Uveal melanoma
Lung cancer
Breast cancer
Clear cell renal cell carcinoma
Melanoma
Osteosarcoma
Prostate cancer
Renal cell carcinoma
Advanced non-small cell lung […]
CRC
Colorectal cancer
Rectal adenocarcinoma
Papillary thyroid carcinoma
Glioblastoma
Non-small cell lung cancer
Pleural mesothelioma
Head and neck squamous cell […]
Pancreatic ductal adenocarcinoma
Bladder cancer
Pancreatic neuroendocrine tumor
INVASIVE ADENOCARCINOMA
Gastric cancer
Cervical cancer
Pituitary neuroendocrine tumor
ADENOCARCINOMA
SCLC
Nasopharyngeal carcinoma
HCC
Part-solid LUAD
High-grade serous ovarian cancer
Esophageal cancer
2
1
0
−1
−2
Median activity (z across cancers)
CM01 CM02 CM03 CM04 CM05 CM06 CM07 CM08 CM09
d
Activity vs. infiltration (• FDR<0.05, |ρ|≥0.2)
SPP1 macrophage
Inflammatory myeloid
Myeloid
CAF
Stromal activation
Cytotoxic
Lymphoid
Immune-hot proxy
Strict malignant
0.4
0.2
0.0
−0.2
−0.4
Within-cancer Spearman ρ (n = 1,303 samples)
CM01 CM02 CM03 CM04 CM05 CM06 CM07 CM08 CM09